

Sensor Cost Report

The final sensor suite of our GNSS-free drone navigator: what it costs, part by part

Team Taipei Drift | EDTH Taiwan Defense Tech Hackathon 2026, Challenge 2

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Summary

This report prices every sensor of our final simulated drone, as specified on the integration branch [46], and compares the optional range finder with LiDAR alternatives. For each sensor the requirement is taken from our simulation's sensor model, matched against commercial parts, and priced from public listings with sources. All prices are single-unit or small-volume list prices, not quotes.

Sensor (final version)	Specification in our simulator	Matching part	List price	Added cost per drone
Down + forward camera	2 × 1024×1024 px, 90° FOV, 25 Hz	Platform camera; reference: Raspberry Pi Global Shutter Camera	\$50 each [36]; Camera Module 3 wide \$35 [14]	\$0 if on the drone; else ≈ \$70–100
IMU	100 Hz, consumer MEMS (Mid-Air noise model)	ICM-42688-P on Pixhawk 6C	flight controller \$211.99 [20][21]	\$0 (on the flight controller)
Barometer	50 Hz, 10 Pa noise (≈ 0.8 m), drift 0.65 Pa/√s	MS5611 on Pixhawk 6C	chip \$5.78–12.32 [15]	\$0 (on the flight controller)
Sun sensor	0.1°, ±65° FOV (Fan et al. 2016)	Self-built: linear array behind a V-slit + RP2040	parts \$6–9 + mechanics (estimate) [9][10][12]	≈ \$10–20
Downward range finder (optional, metric flow)	1 beam, 0.2–100 m, σ 0.02 m, 25 Hz	LightWare LW20/C (0.2–100 m, ±5 cm, 19 g)	\$279 [23][24]	≈ \$250–300
RF direction finder (over water)	AIS receiver; 4 × 17 cm VHF stubs, RF switch at 1 kHz, one radio, −110 dBm	Low-cost: RTL-SDR V4 + ADG904 switch + 4 whips. Reference: HackRF One + Opera Cake	≈ \$110–210 low-cost; ≈ €558 (≈ \$615) reference [37–44]	≈ \$110–210

Table 1. The final sensor suite and its cost. GNSS (1 Hz, used only before the jam) is part of every drone and not costed.

Main findings.

- **The core navigator adds about \$10–20 of hardware.** Cameras, IMU and barometer of the final version are standard on a drone with a Pixhawk-class flight controller [20]. The only new part for camera, sun and map navigation is the sun sensor.

- **The full final suite adds about \$370–530** (sun sensor, LW20/C range finder and a low-cost AIS direction finder), or about \$875–935 with the reference radio hardware of the paper the design follows [37]. Both the range finder and the radio are optional modules: metric flow is off by default, and the radio only helps where AIS ships are in reach.
- **The final range finder specification matches an existing part one-to-one.** The simulator's 0.2–100 m single beam is the LightWare LW20/C's range, at \$279, 19 g, ± 5 cm [24].
- **A space-grade sun sensor buys nothing.** The drone's 1° tilt error, not the sensor, limits the heading (section 3.1). Space units cost €2,500 to \$12,000 [6][7][8].
- **LiDAR navigation is in another price class.** A scanning LiDAR that sees the ground from our flight heights costs about \$2,500 with its computer and weighs about 0.5 kg [30][31][34].

1. Scope and method

1.1 The final sensor suite

The final simulated drone is a Mid-Air-like quadcopter (1 kg, AirSim default airframe) defined in `sim/models/midair_quad/model.sdf` and assembled by `sim/launch/sim.launch.py`; the heading sensor and the camera navigator are configured in `baseline/configs/sim_navigator.yaml` and fused in `scripts/fused_replay.py` [46]. Table 2 lists every sensor as specified there.

Sensor	Specification (integration, d8660c9)	Used by	Where defined
Down camera	1024×1024 px (launch default <code>cam_res</code>), 90° FOV, 25 Hz, pinhole; recordings used 512 px at 5 Hz	Camera navigator, map fixes, ground speed	<code>model.sdf</code> , <code>sim.launch.py</code>
Forward camera	1024×1024 px, 90° FOV, 25 Hz	ESKF vision rotation and direction updates	<code>model.sdf</code>
IMU	100 Hz; noise added by <code>sensor_noise.py</code> : gyro white 0.0005–0.005 rad/s, bias 0.001–0.01 rad/s; accel white 0.005–0.05 m/s ² , bias 0.01–0.1 m/s ²	ESKF prediction	<code>sensor_noise.yaml</code>
Barometer	50 Hz, 10 Pa white noise (≈ 0.8 m, "MS5611-class"), drift 0.65 Pa/ $\sqrt{s}$	ESKF height	<code>model.sdf</code> , <code>sensor_noise.yaml</code>
GNSS	1 Hz, ≈ 1.5 m horizontal, 3 m vertical; cut 20 s after the first fix	Start only	<code>model.sdf</code> , <code>sim.launch.py</code>
Downward range finder	Single beam (<code>gpu_lidar</code>), 0.20–100 m, resolution 0.01 m, σ 0.02 m, 25 Hz, co-located with the down camera	Metric optical flow (<code>metric_flow:=true</code> , off by default)	<code>model.sdf</code> , <code>sim.launch.py</code>
Sun sensor	Digital, Fan et al. 2016: 0.1° , $\pm 65^\circ$ FOV, mounting error per flight (heading.source: <code>sun_digital</code>)	Navigator heading; fused filter yaw	<code>sim_navigator.yaml</code> , <code>sun_sensor.py</code>
RF direction finder	AIS (161.975/162.025 MHz) receiver, sensitivity -110 dBm, NF 6 dB; pseudo-Doppler array: 4 helical stubs, 17 cm, on the arms at 9.5 cm radius, switched at 1 kHz into one radio; bearing σ 3° floor + 1° mounting bias	RF position fix over water (strait world)	<code>rf.yaml</code> , <code>RF_README.md</code>

Table 2. Final sensor suite, as specified on branch `integration` [46].

1.2 How prices were collected

- Public list prices from distributors and vendors, retrieved on 3 October 2026. Where several listings exist, the range is given; quantity breaks are quoted where the source gives them.
- Prices are in the currency of the source. For the per-drone totals we convert at approximately 1 EUR $\approx$ 1.10 USD, 1 GBP $\approx$ 1.30 USD and 1 AUD $\approx$ 0.65 USD. These rates are our assumption for orientation only.
- Items without a public source (slit masks, filters, housings, small PCBs, cables, assembly) are marked as **estimate** and kept separate from sourced prices.
- Not included: export-control and qualification cost, integration labour, taxes and shipping (except where a listing includes VAT, which is stated).

2. Sensors already on the platform

2.1 IMU and barometer

A Pixhawk 6C flight controller (\$211.99 [21]) carries an ICM-42688-P and a BMI055 IMU and an MS5611 barometer [20], the barometer class our simulator names. Any drone flown with such a controller therefore needs no extra IMU or barometer. For a separate barometer:

Part	Key specification (datasheet)	Price	Source
TE MS5611-01BA03	Resolution 0.012 mbar RMS (1.2 Pa) at OSR 4096; 10 cm altitude resolution	\$12.32 (1 pc), \$7.57 (100), \$5.93 (1,000), \$5.78 (reel 3,600)	[15][16]
Bosch BMP390	Relative accuracy ± 3 Pa (± 25 cm); RMS noise 0.02 Pa at lowest bandwidth	from \$5.60 (chip)	[17][18]
Infineon DPS310	Precision ± 0.2 Pa (± 2 cm) in high-precision mode; absolute ± 100 Pa	Adafruit breakout \$6.95 (1–9), \$5.56 (100+)	[19]

Table 3. Barometer candidates.

- All three are quieter than the simulator assumes: the MS5611 by a factor of about 8 (1.2 Pa against 10 Pa). Our simulated results are conservative on barometer noise.
- In flight, weather, temperature and rotor downwash limit the barometer more than the chip does; the simulator's drift term stands in for them.

2.2 Cameras

The final drone has two cameras, down and forward, each 1024×1024 px with a 90° field of view at 25 Hz. Our plan assumes the drone already carries a camera (docs/PLAN.md [46]). For a drone without them, two reference parts: the Raspberry Pi Global Shutter Camera (Sony IMX296, 1456×1088, \$50 [36], lens extra and not priced here) and the Camera Module 3 wide (\$35 [14]). Two cameras: about \$70–100 plus lenses.

3. Sun sensor

3.1 How accurate does it need to be?

The sun sensor measures the sun's direction in the drone's frame; with time and rough position the sun's direction in the world is known, and the difference gives the heading. On a drone two errors add up (baseline/src/sensors/heading.py [46]): the sensor error d , magnified by $1/\cos(e)$ at

sun elevation e , and the IMU's tilt error t , which shifts the sun sideways by $t \cdot \tan(e)$. The heading error is $\sqrt{(d/\cos e)^2 + (t \cdot \tan e)^2}$. With the 1° tilt error our simulator assumes:

Sun elevation	$d = 0.03^\circ$ (miniDSS)	$d = 0.1^\circ$ (Fan et al., our model)	$d = 0.3^\circ$ (uncalibrated miniDSS)	$d = 1.0^\circ$
30°	0.58°	0.59°	0.67°	1.29°
45°	1.00°	1.01°	1.09°	1.73°
60°	1.73°	1.74°	1.83°	2.65°

Table 4. Heading error from a sun sensor on a drone with 1° tilt error, for four sensor accuracies (computed from the formula above).

Consequence: below about 0.3° the sensor no longer matters; the tilt error dominates. A sensor of 0.1 – 0.3° is sufficient. In our simulations the sun heading gave 0.9° median heading error on flight 1 (compass: 1.8° with fixed offsets up to 8° per draw), and in the fused filter it reduced the heading error from 55° to 1.3° (docs/simulation-results.md [46]). The five-photodiode variant was about as good as the compass and failed the 120 m flight, so it is not recommended. The final configuration keeps the digital sensor (heading.source: sun_digital).

3.2 The paper: Micro-Digital Sun Sensor (Xie et al., 2010)

The Micro-Digital Sun Sensor (μ DSS) was developed by TU Delft and TNO for micro-satellites [1][2]. Its core is the APS+, one CMOS chip that holds a 512×512 active pixel sensor (the test chip in the paper: 368×368), a 12-bit ADC, timing, and on-chip centroid computation, made in a standard $0.18 \mu\text{m}$ CMOS process. Sunlight passes through a membrane pinhole onto the array; the sun spot is about 10 pixels wide. To save power the chip first reads one row profile and one column profile (in effect two line sensors) to find the sun, then reads only a small window around it [2].

The concept was carried into the TNO miniDSS prototype [3]: 368×368 pixels of $6.5 \mu\text{m}$ on a 5×5 mm ASIC, 0.03° accuracy (0.3° without unit calibration), 0.01° noise-equivalent angle (3σ), $102^\circ \times 102^\circ$ field of view, output at 0.5–10 Hz, 72 g, 65 mW, designed for low recurrent cost.

Note on "line sensor": the μ DSS uses an area sensor (APS), not a line sensor; only its acquisition mode reads row and column profiles. The sensor in our simulator, Fan et al. (2016), does use a linear detector behind a V-shaped slit [4][5]. Both principles are costed below. **Price of the μ DSS itself:** none; it is a research prototype, as is the miniDSS. The closest prices are those of commercial space sun sensors (Table 5).

Product	Type	Specification	Price	Source
Solar MEMS nanoSSOC-A60	Analog, 2-axis	3.7 g, $27.4 \times 14 \times 5.9$ mm	€2,500	[6]
Lens R&D BiSon64	Analog, space-qualified	catalogue 2018	€4,317 (1–5 pcs); €3,358 in volume	[7]
Lens R&D BiSon64-ET FM	Extended temperature, flight model	catalogue 2018	€8,984 (1–5 pcs); €7,007 (60+)	[7]
NewSpace NFSS-411	Digital fine sun sensor	0.1° RMS, 140° FOV, 5 Hz	\$12,000	[8]

Table 5. Space sun sensors. Their price pays for radiation tolerance, thermal-vacuum qualification and flight heritage, none of which a drone needs.

3.3 Building one from commercial parts

The mask, the neutral-density filter (sunlight saturates a bare sensor; the miniDSS needed an attenuating window [3]) and the housing have no public price; we **estimate** \$2–10 for them together.

Option	Parts	Sourced price	Estimate (mask, filter, housing)	Total per unit
A. Line sensor + V-slit (Fan et al., our simulated model)	ams TSL1401CL 128×1 linear array; RP2040 microcontroller	TSL1401CL \$5.49–8.06 (1 pc), \$2.50–3.71 (1,000) [9][10]; RP2040 \$0.70–1 [12]	\$2–10	≈ \$9–19 (1 pc); ≈ \$5–15 in volume
A+. Same, higher-grade line sensor	Hamamatsu S11639-01 (2,048 px)	€224.72–231.67 [11]	\$2–10	≈ \$250–265
B. Pinhole + area sensor (μDSS principle)	OV9281 1 MP global-shutter module; processing on the companion computer	≈ £25 (≈ \$32) [13]	\$2–10	≈ \$34–42
C. Upward camera (Alessandro's sun detector)	Raspberry Pi Camera Module 3	\$25–35 [14]	\$2–10 (filter)	≈ \$27–45

Table 6. Sun sensor built from commercial parts. Option A is the cheapest and is the type our simulator models.

- **Accuracy is not yet shown.** Fan et al. reached 0.1° and the miniDSS 0.03° in their laboratories [3][4]. A build from commercial parts has not been tested by us; the requirement from Table 4 ($\leq 0.3^\circ$) leaves a margin of three to ten times.
- **Mass and power.** Fan et al.: 35 g, 200 mW [4]; miniDSS: 72 g in a space housing, 65 mW [3]. A drone build should be lighter; we have not weighed one.
- **Operational limits** (from our model): no heading when the sun is hidden by cloud (the gyro carries the heading), outside the field of view, or near the zenith (in Taiwan around midday from May to July). The magnetometer on the flight controller (IST8310 [20]) remains the fallback.

4. Downward range finder and LiDAR

4.1 The final range finder

The final drone carries one narrow downward beam next to the down camera: 0.20–100 m, 1 cm resolution, σ 0.02 m, 25 Hz (model.sdf; launch options range_min_m, range_max_m, range_noise_std_m [46]). It scales the down camera's optical flow into a metric ground speed. The measurement is off by default (metric_flow:=false). Over flat ground (terrain world, 39 m) its speed readings were 0.16 m/s off in the median and the filter held 14.5 m median for 164 s, against 322 m without it; over the city (buildings 16–63 m, flight at 80 m) the speed was 4.45 m/s off because range and tracked points lay at different depths (docs/simulation-results.md [46]).

Note: a 100 m limit does not reach the ground from our 110 m and 120 m Wufeng flights. If the range finder is to serve those heights, the limit must be raised and a longer-range part chosen (Table 7, last rows of the single-beam group).

4.2 Candidates

Part	Type	Range	Accuracy / mass	Price	Matches final spec?	Source
Benewake TF02-Pro	single beam	40 m	— / 50 g	€94–120	no (range)	[22]

Part	Type	Range	Accuracy / mass	Price	Matches final spec?	Source
LightWare LW20/C	single beam	0.2–100 m	±5 cm / 19 g	\$279	yes, one-to-one	[23][24]
Benewake TF03-180	single beam	100 m operating, 180 m max	— / —	\$249.90–274.95	yes (range)	[25]
LightWare SF30/C	single beam	100 m	— / —	\$299	yes (range)	[23]
LightWare SF30/D	single beam	200 m	— / 36 g	\$399	yes, also above 100 m	[23]
LightWare GRF-500	single beam	0.2–500 m	— / 10.7 g, 0.6 W	€524.95 incl. VAT (≈ €441 net)	yes, also above 100 m	[26][27]
Livox Mid-360	scanning	40 m at 10 % reflectivity	— / 265 g	≈ \$1,092	no (range)	[28][29]
Livox Avia	scanning, triple return	up to 450 m	— / 498 g	\$2,079 (US); AUD 4,290	different sensor class	[30][31]
Ouster OS1-32	scanning	—	—	\$5,770 (eBay) – \$8,000 (2019 list)	different sensor class	[32]
Jenoptik DLEM 20	laser range finder module	up to 5 km	±0.5 m / < 33 g	not published	TRN class	[33]

Table 7. Range finder and LiDAR candidates. "—" = not taken from a source in this report.

- **Recommended:** LightWare LW20/C, \$279: its range is exactly the simulator's 0.2–100 m, ±5 cm is comparable to the simulated σ 2 cm, and it reads 48 to 5,000 times a second, well above the 25 Hz used [24]. Cheaper with the same reach: TF03-180 at \$250–275 [25]. For flights above 100 m: SF30/D (\$399) or GRF-500 (≈ €441 net).
- **Scanning LiDAR** (LiDAR-inertial odometry) is not part of our design. The affordable Mid-360 does not see the ground from 65–120 m; the Avia does, at ≈ \$2,500 including a Jetson Orin Nano Super (\$399 [34]) and about 0.5 kg.
- **Felix's laser TRN** (σ 0.3 m, 10 Hz, up to 5 km, three echoes; TRN/configs/laser.yaml) needs a km-class module such as the DLEM 20; its price must be requested.
- **Every LiDAR emits:** 905 nm pulses are visible to night-vision and near-infrared cameras and to laser warning receivers. The cameras and the sun sensor are passive; the AIS receiver only listens.

5. RF direction finder (AIS, over water)

5.1 The final design

Over open water the camera has nothing to match. The final drone then takes bearings to ships' AIS transmitters (161.975 and 162.025 MHz) with a pseudo-Doppler direction finder: four 17 cm helical VHF stubs on the rotor arms (square of 13 cm side, 0.07 wavelength), switched one after another into a single receiver 1,000 times a second; the phase of the resulting Doppler tone gives the bearing. Sensitivity –110 dBm, noise figure 6 dB, bearing σ 3° plus a 1° mounting bias (sim/config/rf.yaml, sim/RF_README.md [46]). The design follows Gerhard and Tokekar (2020), who flew four antennas, an Opera Cake switch board and a HackRF One on a four-rotor UAS at 150 MHz [37]. In the latest strait-world run the ships' bearings alone gave 55 m median error over 3.5 minutes (docs/simulation-results.md [46]).

5.2 Cost

Item	Reference build (as in [37])	Low-cost build	Source
Receiver	HackRF One, €305	RTL-SDR Blog V4, \$39.95 (with antennas, launch price) – \$84.95 (SparkFun kit)	[39][40][41]
RF switch	Opera Cake, €189 / US\$190	Analog Devices ADG904 SP4T, \$5.88 (1 pc), \$3.30 in volume	[38][39][43]
Switch timing	Opera Cake firmware	RP2040 microcontroller, \$0.70–1	[12]
4 antennas	4 × Nagoya NA-701 (22 cm VHF/UHF whip), €15.95 each	4 × Nagoya NA-701, \$14.99–24.99 each	[44]
PCB, coax, mounts	—	estimate \$5–15	—
Total	≈ €558 (≈ \$615)	≈ \$110–210	

Table 8. Cost of the AIS direction finder, as a copy of the published hardware and as a low-cost build.

- **Antenna caveat.** The simulator assumes 17 cm helical stubs. The NA-701 is a 22 cm dual-band whip; marine AIS helical stubs of 15 cm exist but are sold with brackets and 20 m of cable at \$85.49 or more each [45], which would add about \$240–280 for four.
- **Receiver caveat.** The low-cost build relies on the switching being synchronised with the RTL-SDR's samples; the published system used the HackRF and the Opera Cake [37]. It has not been built or tested by us.
- **Alternative:** a coherent five-channel receiver (KrakenSDR, \$499 [42]) measures bearings by interferometry instead of switching. It costs more and is larger than the switched design.

6. Cost per drone: configurations compared

Configuration	Added hardware	Added cost (≈ USD)	Share of a \$500 drone
1. Core navigator (cameras, IMU, barometer on the drone)	Sun sensor, option A	\$10–20	2–4 %
2. Final full suite, low-cost radio	Sun sensor + LW20/C (or TF03) + low-cost AIS direction finder	\$370–530	74–106 %
2b. Final full suite, reference radio	as 2, radio as in Gerhard and Tokekar	\$875–935	175–187 %
3. Platform parts, if the drone has none	Pixhawk 6C + 2 cameras + Raspberry Pi 5 (4 GB)	\$367–397 (plus lenses)	73–79 %
4. Scanning LiDAR instead of the range finder	Livox Avia + Jetson Orin Nano Super	≈ \$2,480	≈ 500 %
5. High-end scanning LiDAR	Ouster OS1-32 + Jetson	\$6,170–8,400	1,230–1,680 %
6. Space-grade sun sensor instead of option A	nanoSSOC-A60 ... NFSS-411	\$2,750–12,000	550–2,400 %; no accuracy benefit (Table 4)

Table 9. Added hardware cost per drone, from Tables 1–8. Conversions at the rates in section 1.2; the \$500 airframe is the orientation agreed with our mentor (docs/PLAN.md [46]).

7. Conclusions and recommendations

1. **Core navigator:** about \$10–20 of new hardware, the sun sensor. Cameras, IMU and barometer of the final version are standard on a Pixhawk-class drone.

2. **Sun sensor:** build option A (TSL1401CL behind a V-slit, read by an RP2040) and bench-test it against a known sun direction; the requirement is $\leq 0.3^\circ$. Do not buy a space-grade sensor.
3. **Range finder:** if metric flow is used, the LightWare LW20/C (\$279) matches the final specification; for flights above 100 m use the SF30/D or GRF-500.
4. **AIS direction finder:** a low-cost build is about \$110–210 but untested; the published reference hardware costs about \$615. Build and bench-test one before quoting a number to customers.
5. **For the pitch:** the core adds about \$10–20 per drone; the complete final suite about \$370–530; a scanning LiDAR alone about \$2,500. The sensors are passive except the optional range finder.

8. Limitations

- List prices of one day; distributor prices, stock and exchange rates change. Defence volumes and export-controlled items are priced on request.
- Mechanics, PCBs, cables and assembly of the self-built sun sensor and direction finder are estimates, not sourced prices.
- The accuracy of the self-built sun sensor and of the low-cost direction finder is assumed from the literature, not measured by us.
- Specifications are as published by manufacturers or retailers; we did not test the parts. Some secondary listings (eBay, regional retailers) were used where no primary price was available; these are marked by source.
- The specifications come from the simulator; they are what our results assume, not a flight-qualified design.

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baseline/src/sensors/heading.py and sun_sensor.py; scripts/fused_replay.py; TRN/configs/laser.yaml;
docs/simulation-results.md; docs/PLAN.md. <https://github.com/dwn97/TaipeiDrift/tree/integration>
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